



Git

Hands-On 1

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Description:

This hands-on lab introduces foundational Git commands—such as `git init`, `git status`, `git add`, `git commit`, `git push`, and `git pull`—while guiding you through setting up your environment, integrating Notepad++ as your default editor, and managing your source code repository.

WorkFlow

Step 1: Setup Machine with Git Configuration

Verify Git Installation: Open the Git Bash shell and check that the Git client is installed properly by running:

`git version`

Set User-Level Configuration: Configure your global user name and email address:

`git config --global user.name "username"`

`git config --global user.email "username@cognizant.com"`

Verify Configuration: Confirm your settings are correct by listing the global configurations:

`git config --global --list`



WorkFlow

Step 2: Integrate Notepad++.exe to Git and Make It the Default Editor

Add Path to Environment Variables: If Git Bash returns a command not found error for notepad++, add the path of notepad++.exe to your Windows System Environment Variables (Path variable under User/System variables via Control Panel -> System -> Advanced System settings).

Configure Notepad++ Alias: Open a fresh Git Bash shell and create an alias profile by adding the following line using Notepad++:

```
alias npp='notepad++.exe -multinst -nosession'
```

Set Core Editor: Set Notepad++ as Git's global default editor:

```
git config --global core.editor "notepad++.exe -multinst -nosession"
```

Verify Default Editor: Check your configuration settings to ensure Notepad++ is successfully set as the editor:

```
git config --global -e
```



WorkFlow

Step 3: Add a File to Source Code Repository

Initialize Repository: Create and initialize a new project folder named GitDemo:

git init GitDemo

Check Working Directory: View hidden files and verify the .git directory:

ls -al

Create and Verify a File: Create a file named welcome.txt with initial content, check that it exists, and inspect its contents:

echo "Welcome to the version control" >> welcome.txt

ls -al

cat welcome.txt

Check Status: View the repository status to see untracked files:

git status

Stage and Commit: Stage the file for tracking and commit your changes using the default editor for multi-line comments:

git add welcome.txt

git commit

Final Status Check: Verify that the working tree is clean and the file is safely committed to the local repository:

git status

Sync with Remote Repository: Pull updates from or push your local GitDemo repository to your remote repository hosted on GitHub/GitLab:

git pull origin master

git push origin master



Git Bash & Output

```
MINGW64:/c/Users/AADI

AADI@Aadi MINGW64 ~
$ git --version
git version 2.55.0.windows.3

AADI@Aadi MINGW64 ~
$ git config --global user.name "Aadi4-gif"

$ git config --global user.email "krithn.aadityan@gmail.com"

AADI@Aadi MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=krithn.aadityan@gmail.com
user.name=Aadi4-gif

AADI@Aadi MINGW64 ~
$
```

```
MINGW64:/c/Users/AADI

AADI@Aadi MINGW64 ~
$ notepad++

AADI@Aadi MINGW64 ~
$ git config --global core.editor "notepad++.exe -multiInst -nosession"

AADI@Aadi MINGW64 ~
$ git config --global -e
hint: Waiting for your editor to close the file...
```

```
MINGW64:/c/Users/AADI/GitDemo

drwxr-xr-x 1 AADI 197121 0 Jul 24 13:14 ../
drwxr-xr-x 1 AADI 197121 0 Jul 24 13:14 .git/
-rw-r--r-- 1 AADI 197121 31 Jul 24 13:15 welcome.txt

AADI@Aadi MINGW64 ~/GitDemo (master)
$ cat welcome.txt
welcome to the version control

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    welcome.txt

nothing added to commit but untracked files present (use "git add" to track)

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git add welcome.txt
warning: in the working copy of 'welcome.txt', LF will be replaced by CRLF the next time Git touches it

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git commit
hint: Waiting for your editor to close the file...
```

```
MINGW64:/c/Users/AADI/GitDemo

AADI@Aadi MINGW64 ~
$ cd GitDemo

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git add welcome.txt

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git commit -m "Welcome"
[master (root-commit) 088c41c] Welcome
1 file changed, 1 insertion(+)
create mode 100644 welcome.txt

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 247 bytes | 247.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:   https://github.com/Aadi4-gif/GitDemo/pull/new/master
remote:
To https://github.com/Aadi4-gif/GitDemo.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git remote -v
origin https://github.com/Aadi4-gif/GitDemo.git (fetch)
origin https://github.com/Aadi4-gif/GitDemo.git (push)

AADI@Aadi MINGW64 ~/GitDemo (master)
$ AC

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git pull origin master
From https://github.com/Aadi4-gif/GitDemo
 * branch            master       -> FETCH_HEAD
Already up to date.

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git push origin master
Everything up-to-date

AADI@Aadi MINGW64 ~/GitDemo (master)
$ AC

AADI@Aadi MINGW64 ~/GitDemo (master)
$ |
```




Git Bash & Output

```
C:\Users\AADI\gitconfig - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
.gitconfig
1 [user]
2   email = krithn.aadityan@gmail.com
3   name = Aadi4-gif
4 [core]
5   editor = notepad++.exe -multiInst -nosession
6
```

```
C:\Users\AADI\GitDemo\git\COMMIT_EDITMSG - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
COMMIT_EDITMSG
1
2 # Please enter the commit message for your changes. Lines starting
3 # with '#' will be ignored, and an empty message aborts the commit.
4 #
5 # On branch master
6 #
7 # Initial commit
8 #
9 # Changes to be committed:
10 #   new file:   welcome.txt
11 #
12
```



Git Bash & Output

